

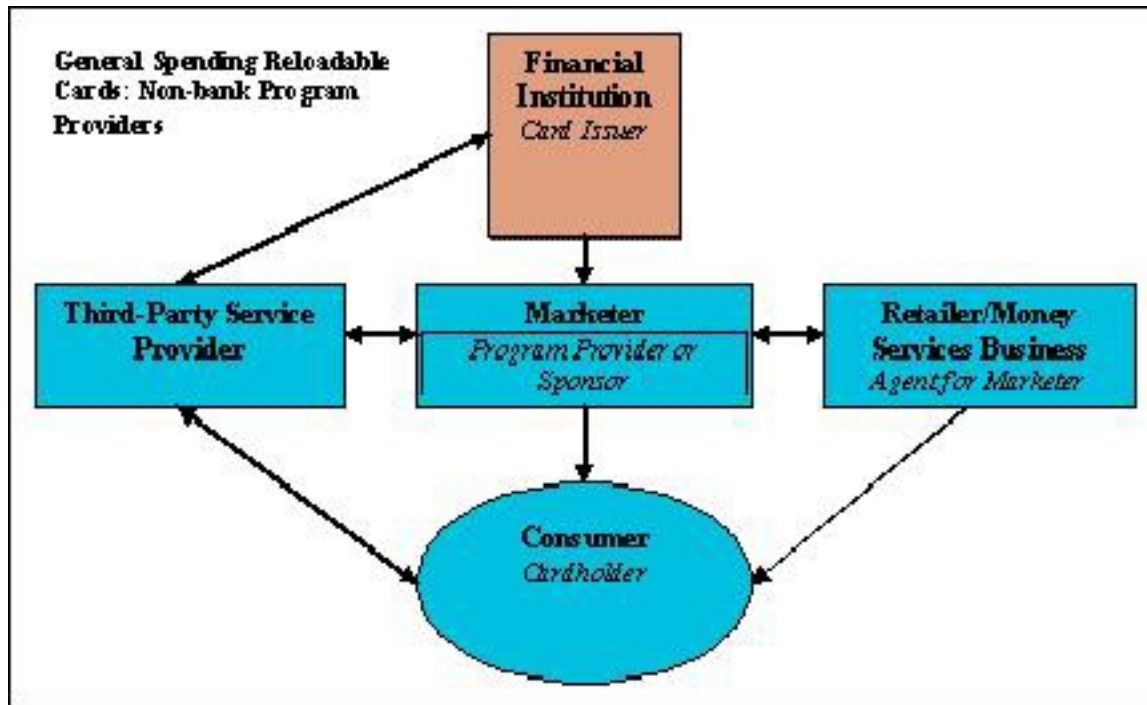
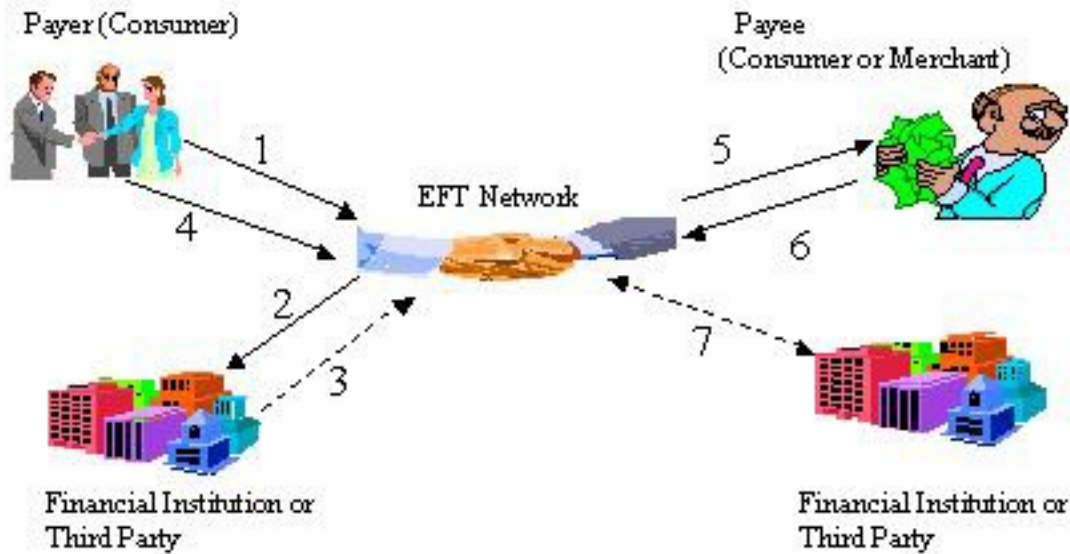


Figure 9: General spending card program offered by nonbank providers

Prepaid card transactions typically follow the "four corner" pattern in Figure 10. The consumer purchases a prepaid card (Step 1 and Step 2). When the consumer pays for goods or services with the card, electronic notations or tokens transfer from the card to the merchant's cash register (Step 3, Step 4, and Step 5). The merchant contacts the computer network of the financial institution that issued the prepaid card and presents the tokens for payment (Step 6). The network notifies the consumer's financial institution to pay the appropriate sum to the merchant's financial institution, and net settlement occurs at the end of the business day (Step 7). The financial institution keeps a percentage of the payment (the discount) as compensation for the services provided.



Legend: Solid lines represent the flow of information and dashed lines represent the flow of funds.

Figure 10: Stored Value Card Clearing and Settlement

There are many configurations of third parties and financial intermediaries, and there is a significant number of prepaid cards in circulation for which the four-corner diagram is not sufficient. The financial intermediary may hold the funds supporting the circulating stored value in a pooled account, with a third-party keeping the record of the individual transactions. Financial businesses that are not traditional financial institutions may be the issuers and may distribute the cards through retailers.

If the prepaid card is not a smart card, the associated funds are kept in a separate account. When a customer uses the prepaid card, the merchant sends a message to the record-keeping entity to determine whether the balance is sufficient to cover the transaction. If funds are available, the third party or financial institution processes the transaction.

This account arrangement may be used for smart cards also, with the accounts debited when the merchant presents tokens for payment. Although financial institutions issue prepaid cards and maintain account records, third parties may be involved in maintaining individual account records also.

Three general-spending prepaid card programs that increasingly are offered by financial institutions include branded remittance cards, teen cards, and gift cards.

Remittance Cards

With the growing demand for global person-to-person money transactions, an increasing number of bank-issued cards are being used to make remittances. In many cases, the sender of the remittance lives in the U.S. and uses a financial institution to electronically transfer money to a pre-established, branded prepaid card account. A financial institution in the sender's or recipient's country issues a prepaid card to the recipient. The recipient can use the card to obtain cash at an ATM or goods and services at a merchant POS. Alternatively, the sender may use a branded prepaid card to send funds to a recipient via the Internet. The recipient receives the funds either in cash or in credits made to an existing prepaid card account or a bank account.

Teen Cards

Another stored-value product gaining favor among consumers is the teen card that is marketed to help parents instill financial responsibility in their children while monitoring and supervising their spending. The consumer typically funds the prepaid card with the issuing financial institution through a withdrawal from a deposit account or by charging a credit card.

Gift Cards

Gift cards were initially offered by retailers as a replacement for paper-gift certificates and operated in closed-loop payment systems. In recent years, financial institutions noted the rising popularity and market potential and included gift cards in their product offerings thereby competing with retailers. Gift cards issued by a financial institution typically are card network branded and operate in an open-loop payment system, making them functional at ATMs and at any POS that accepts network debit and credit cards.

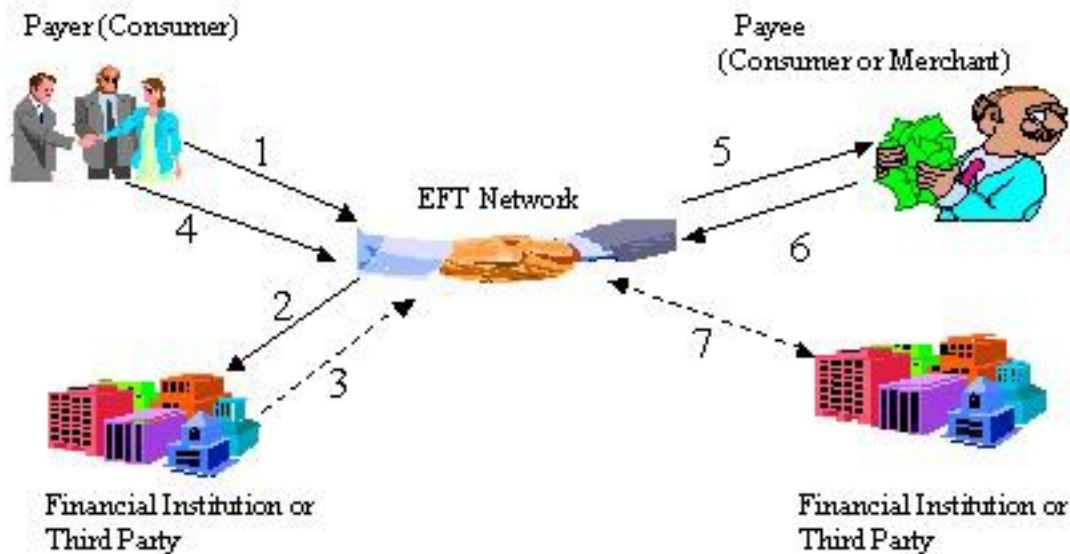
Online Person-to-person (P2P), Account-to-Account (A2A) Payments and Electronic Cash

Other electronic payments include person-to-person, account-to-account, electronic cash, and electronic benefit transfers. These payment instruments are usually associated with an established consumer deposit account and facilitate consumer access to recurring or one-time debit and credit transactions and a variety of federal, state, and local government benefit programs.

Online P2P or e-mail payments typically use traditional payment networks to transfer funds electronically from one consumer to another. Though these payments are named for their ability to send funds among individuals online, the majority of P2P payments are Internet purchases at online auctions or small businesses. In most cases, P2P transfers use existing retail payment systems to add and withdraw funds from accounts. The simplest case is when the person making a payment and the receiver maintain accounts at the same bank. This type of payment is called an "on-us" transaction. They are settled by posting accounting entries on the books of one financial institution. P2P transfers also may occur outside the traditional payment networks and, in their simplest form, may take place as an exchange of cash between two individuals. As technology advances, the transfer of funds through the use of proximity devices, such as mobile telephones and personal digital assistants (PDAs), is likely.

Most P2P services charge to the receiver of the funds a fee that varies depending upon

various factors, including payment method and the sender's credit history. Payments made with funds that originated from either ATM or ACH transactions are less expensive than payments made with funds originated from credit cards. P2P systems may offer to the receiver an opportunity to obtain funds through a check and for an additional fee.



Legend: Solid lines represent the flow of information and dashed lines represent the flow of funds.

Figure 11: Online P2P Clearing and Settlement

Online P2P payments typically occur using the process described in Figure 11. The sender of the funds must have an account with the P2P service provider (Step 1). Depending upon the service, the funds may come from an existing credit card or transaction account or may be drawn from a previous balance with the online P2P payment provider (Step 2 and Step 3). The sender can designate the e-mail address of the intended funds recipient (Step 4). The P2P network transfers the funds to the receiver's account as an "on-us" transaction. Once the funds reach the receiver's account, notice of the transaction is sent through e-mail to the receiver (Step 5). The receiver of the funds must join the service if it does not already have an account (Step 6). The online P2P payment service can disburse the funds from the receiver's P2P account through an ACH payment, a check payment, an EFT credit, prepaid card, or a credit to a credit card account (Step 7).

Account-to-account (A2A) payments are similar to P2P payments. They involve the transfer of funds from one customer's account to another account at either the same or another financial institution. Like P2P payments, A2A transfers can be initiated through the customer's Internet banking service, a biller's payment Web site, or by telephone instruction from the customer. Unlike P2P transfers, consumers must access an existing

retail payment account (deposit account) at a financial institution in an A2A transaction. To complete a transaction, the customer must know the recipient's account number or some other identifier. A2A payments can be effected on the ACH or ATM networks. On the ACH networks, funds are cleared and settled within two to three days. The ATM networks may allow same-day funds availability although settlement may not occur for two or three days. Same-day transfers using the ATM networks are usually less expensive than traditional wire transfers.

P2P payments are a growing segment of the A2A market. The success of the P2P online auction model is attributed to the consumers' demand for convenient and reliable P2P transactions. P2P payments may include transaction accounts and may be conducted through the use of proximity devices such as mobile telephones or PDAs. P2P payments are expected to grow as more reliable and convenient payment methods are introduced.

Financial institutions and retailers are also developing electronic cash-payment instruments. Similar to P2P payments, individuals can transfer electronic cash value to other individuals or businesses, generally through the Internet. Consumers can use the cash payment instruments for purchases at retailers' Web sites or they can transfer cash to other individuals through e-mail. Pre-funded accounts that consumers can use for online auction payments are among the most recent applications. In these applications, individuals use a credit card or signature-based debit card number to pre-fund the Web certificate or electronic account, and recipients redeem the value from the issuer.

Electronic Benefits Transfer (EBT)

EBT systems allow recipients of government benefits to authorize transfers from their benefits accounts to health care providers and retailers. The federal government and several states routinely use these accounts to issue food stamps and other benefits. The government distributes all food stamp benefits using this technology and, while the average transaction value is low, total transaction volumes are significant. The institution holding the account authenticates transactions using PIN technology. EBT programs now use cards with either magnetic-strip or microchip technology. Since cards using chip technology have larger storage capacities than cards with a magnetic strip, they can handle more complex transactions. Security measures can be encoded on the card strip or microchip as well to help prevent unauthorized use.

Emerging Retail Payment Technologies

This section discusses several emerging retail payments technologies that financial institutions are implementing or considering. The success of emerging retail payment methods depends upon four key drivers: reliability, cost, convenience, and speed. In terms of the preferences by consumers, merchants, and payment processors, the key drivers are technological advances, convenience, and lower transaction costs. The evolution of such preferences is facilitated by traditional financial institution relationships and established payments networks and infrastructure. Internet, mobile, and contactless payments may be used alone or together to facilitate electronic transactions, further reducing the use of paper checks. The use of currency is expected to retain some appeal because of its anonymity; however, the substitution of electronic payment vehicles for cash micro payments (transactions under \$5.00) is expected to increase.

While the environment for emerging payments is highly dynamic, the most important

emerging payments today are electronic bill presentment and payment (EBPP), P2P, A2A, and stored-value instruments. Several more recent emerging payment mechanisms are contactless payments, biometrics, and proximity payments as well as the format and transmission mechanics used to effect these payments.

Contactless Payment Cards, Proximity Payments and Other Devices

Contactless cards and key fobs have an embedded computer chip with financial and personal information used for payment transactions, and they employ RFID technology for payment transmission. The contactless cards include a microcontroller (or equivalent intelligence) and internal memory and have the ability to secure, store, and provide access to data on the card. The microcontroller also supports the use of improved security features including authenticated information access and information privacy. Traditional plastic cards are easily transitioned to these new contactless cards. Other smart-card technologies provide similar capabilities but do not have the radio frequency interface that would enable them to be read quickly and conveniently at a short distance from the reading mechanism.

Proximity payments are POS transactions made with a mobile device like a cellular telephone, smart card, PDA, or virtually any device that can house a microchip. If the payment is executed with a mobile phone, it may be referred to as an M-payment. Proximity payments are faster, cheaper, and easier than traditional payment mechanisms such as cash or credit card type transactions, particularly for micro payments. Many of these transactions use the same credit/debit card network, and provide lower costs to institutions and to merchants.

Proximity payments and contactless cards permit the consumer to maintain physical control of the access device rather than relinquishing such control to an operator at a POS. Bankcard companies and governmental agencies have become the leaders in facilitating these transactions. Currently, there are multiple transmission types in use, and several are discussed below. Other transmission types are undergoing market test trials.

Financial institutions offering advanced payment technologies (i.e., commercial POS systems to merchants or consumer proximity devices) need to perform the same due diligence and vendor management as they would on any service provider. This includes ensuring an appropriate level of security in the devices.

Biometrics for Payment Initiation and Authentication

Biometric payment services allow a consumer to make purchases or to cash checks using a biometric identifier such as a finger scan linked to his or her personal identification information, accounts at a financial institution, or loyalty programs. Other biometric methods include voice scanning and iris and retinal imaging. Biometric technologies are used increasingly for consumer account authentication. However, a biometric identifier alone is only a single factor, and it may need to be combined with other technologies or factors for proper authentication of high-risk banking transactions.

^[36] FFIEC Guidance "Authentication in an Internet Banking Environment - Supplement," June 2011 www.ffiec.gov/press/pr062811.htm As new payment systems emerge, industry demands for anti-fraud measures may result in greater use of biometrics.

Emerging Network Technologies

The previously discussed emerging payment systems rely upon, and may be integrated with, underlying network communication technologies and protocols. If not properly implemented, new and emerging network communication technologies may expose the payment device or system to additional vulnerabilities. This is particularly true with any network that relies upon broadcast technology to send and receive information. Even close proximity wireless devices, such as RFID, have been found to be vulnerable to eavesdropping at distances greater than they were designed for. Care should be taken to ensure that the underlying network communication technology has security appropriate to the information being transmitted. Currently, there are four types of short-range wireless connectivity technologies that can be used to connect payment devices to POS devices. These include: Infrared, RFID, NFC, and Bluetooth.

Infrared

Infrared communication technology works similarly to a television remote control as information is sent from a device to a payment terminal via a frequency that is invisible to the naked eye. These devices can have signals that are stronger than other contactless technologies and can work from several yards away. Security concerns arise regarding the ability to compromise a transmission because of the strength of the signal. This concern is somewhat mitigated because there must be a direct line of sight for the transmission to work. The Infrared Financial Messaging Group (IrFM) is a consortium of technology and financial companies (including Visa) that work together to promote uniform and interoperable standards ^[37] for infrared devices. These standards include encrypted channels.

Radio Frequency Identification

RFID is a method of remotely storing and accessing data on devices called RFID tags/transponders. An RFID tag can be incorporated into a plastic card (as with contactless cards), a fob, or other device. RFID tags also can be embedded into any product to track inventory. RFID tags contain antennas that enable them to communicate via radio frequency with an RFID transceiver. The technology protocol most widely used for RFID is the ISO 14443 standard. This standard is very general and can be used for multiple types of media and a broad range of hardware.

Near Field Communication

NFC is another short-range communication technology similar to RFID, but based on the ISO 18092 standard. NFC chips can be embedded in a mobile device such as a telephone to enable it to act as a contactless payment card. NFC has additional functionality such as the ability to act as a reader of other NFC devices, thus enabling two consumer devices to share data or transact payments with each other. NFC chips can also be integrated with other applications within the mobile device to permit transactions from multiple accounts.

RFID and NFC have become very flexible solutions for alternative payments. Financial institutions are adding RFID tags to credit and debit cards to speed transactions. In some parts of the world, consumers can link their credit or debit accounts to cell phones enabled with RFID or NFC technology to make purchases at retail sites equipped with payment readers.

Bluetooth

Bluetooth is a close-range wireless radio frequency communication protocol that has been implemented in a wide range of technologies. Bluetooth uses a stronger signal than RFID or NFC and is detectable at greater distances. There has been limited adoption of this protocol.

Retail Payment Systems Risk Management

Action Summary

Financial institutions engaged in retail payment systems should establish an appropriate risk management process that identifies, measures, monitors, and limits risks.

Management and the board should manage and mitigate the identified risks through effective internal and external audit, physical and logical information security, business continuity planning, vendor management, operational controls, and legal measures.

Risk management strategies should reflect the nature and complexity of the institution's participation in retail payment systems, including any support they offer to clearing and settlement systems. Management should develop risk management processes that capture not only operational risks, but also credit, liquidity, strategic, reputational, legal, and compliance risks, particularly as they engage in new retail payment products and systems. Management should also develop an enterprise wide view of retail payment activities due to cross-channel risk. These risk management processes should consider the risks posed by third-party service providers.

Financial institutions should tailor their risk management strategies to the nature and complexity of their participation in retail payment systems, including any support they offer to clearing and settlement systems. Financial institutions must comply with federal and state laws and regulations, as well as with operating rules of clearing houses and bankcard networks. From the initiation of a retail payment transaction to its settlement, financial institutions are exposed to certain risks. For individual retail payment transactions, risks resulting from compliance issues and potential operational failures including fraud are always present. Operational failures can increase costs, reduce earnings opportunities, and impair an institution's ability to reflect its financial condition accurately. Participation in retail payment systems may expose financial institutions to

credit, liquidity, and operational risk, particularly during settlement activities. In addition, a financial institution's credit, liquidity, and operational risks may be interdependent with payment system operators and third parties.

Risk profiles vary significantly based on the size and complexity of the financial institution's retail payment system products and services, IT infrastructure, and dependence on third parties. All financial institutions should maintain an effective internal control environment commensurate with the level of retail payment products and services offered. Effective internal controls should include financial, accounting, technical, procedural, and administrative controls necessary to minimize risks in the retail payment transaction, clearing, and settlement processes. These measures reduce operational and credit risks, ensure individual transactions are valid, and mitigate processing and other errors. Effective controls also ensure supporting IT and network infrastructure promote retail payment transaction integrity, confidentiality, and availability. Financial institutions engaging in retail payment system services should be aware of the risks inherent in the activity.

Financial institutions have always offered a variety of retail payment services; however, recent technological advances are expanding the opportunities for the development of innovative payment products and services. Financial institutions should recognize the reputation and strategic risk of newer products and services, which may lack consumer acceptance. Often, participants will also face uncertainty regarding how state and federal laws and regulations will apply to new payment systems. The ongoing shift from paper to electronic payments is increasing the participation of nonbanks in various payment functions, such as payment processing. Financial institutions should have a comprehensive and effective vendor and third-party service provider risk management and oversight program. ^[38]

Payment System Risk (PSR) Policy

Payment and securities settlement systems are critical components of the nation's financial system. The smooth functioning of these systems is vital to the financial stability of the U.S. economy. The Federal Reserve Board has developed the PSR policy to address risks that payments and securities settlement systems present to the financial system and to the Reserve Banks.

The Reserve Banks are exposed to credit risk when they process wholesale and retail payments for financial institutions holding reserve accounts, just as financial institutions assume credit risk when offering retail payments to their customers. Part of the Federal Reserve's PSR Policy seeks to control and reduce credit risk to the Reserve Banks by controlling financial institutions' use of Federal Reserve daylight overdrafts.

A daylight overdraft occurs when there are insufficient funds in a financial institution's Federal Reserve account to cover the institution's payment activity, such as outgoing Fedwire® funds transfers or ACH credit originations, as outgoing payments are posted during the day.

To control daylight overdrafts, the PSR policy establishes limits, or net debit caps, on the amount of Reserve Bank daylight credit that a depository institution may use during a single day and over a two-week reserve maintenance period. These limits are determined jointly through assessments by the depository institution and its Reserve Bank. The limits reflect the overall financial condition and operational capacity of each

institution using Reserve Bank payment services.

Financial institutions may be monitored on an ex post (i.e., end of day) or real-time basis. Under the Federal Reserve's ex post monitoring procedures, an institution with a daylight overdraft in excess of its maximum daylight overdraft capacity or net debit cap may be contacted by its Reserve Bank. The Reserve Bank may counsel the institution and discuss ways to reduce its excessive use of intraday credit. Each Reserve Bank retains the right to protect its risk exposure from individual institutions by unilaterally reducing net debit caps, imposing collateralization or clearing balance requirements, rejecting or delaying certain transactions, or, in extreme cases, taking the institution off-line or prohibiting it from using Fedwire. In addition, the Reserve Banks assess fees for daylight overdrafts above a certain deductible amount. ^[39] A Reserve Bank will monitor an institution's position in real time when the Reserve Bank believes that it faces excessive risk exposure, for example, from institutions with chronic overdrafts in excess of what the Reserve Bank determines is prudent. In addition, the Reserve Bank will reject or delay certain transactions that would exceed the institution's maximum daylight overdraft capacity or net debit caps, and take other prudential action, including requiring collateral.

Institutions that are monitored in real time must fund the total amount of their ACH credit originations in order for the transactions to be processed by the Reserve Bank, even if those transactions are processed one or two days before settlement. ^[40]

The financial institution's board of directors is responsible for PSR policy compliance and should ensure that management establishes sound internal operating practices, including compliance with applicable banking laws, and carefully manages retail payment system-related financial risks. At a minimum, a financial institution's board of directors and senior management should:

- Understand the financial institution's practices and controls regarding the risks of processing transactions for both its own account and the accounts of its customers and respondents;
- Manage its Federal Reserve account effectively and use daylight credit prudently in accordance with the PSR policy;
- Establish prudent limits on the daylight overdraft or net debit position in its Reserve Bank reserve account and any private-sector clearing and settlement system; and
- Review periodically the institution's daylight overdraft activity to ensure the institution operates within the established guidelines.

Strategic Risk

Strategic risk is associated with the financial institution's mission and future business plans. This risk category includes plans for entering new business lines, expanding existing services through mergers and acquisitions, and enhancing infrastructure (e.g., physical plant and equipment, IT, and networking). The variety of emerging technologies for retail payments demands integration of payment strategies into the financial institution's overall strategic planning processes. Financial institutions also compete increasingly with highly innovative nonbank entities to provide retail payment services.